```
r1.lower.middle, step 2 (2026-11-02)

 chain     seed iterations.saved acceptance.rate final.log.likelihood best.log.likelihood errored.proposals
     1 20263016             5000           0.276             -38396.0            -38373.1                 0
     2 20263017             5000           0.277             -38386.4            -38378.8                 0
     3 20263018             5000           0.277             -38391.6            -38379.1                 0
     4 20263019             5000           0.275             -38396.3            -38378.6                 0
                                                file
 mcmc_r1.lower.middle_step2_chain1_2026-10-30.Rdata
 mcmc_r1.lower.middle_step2_chain2_2026-10-31.Rdata
 mcmc_r1.lower.middle_step2_chain3_2026-10-31.Rdata
 mcmc_r1.lower.middle_step2_chain4_2026-11-01.Rdata

Worst R-hats (> 1.1 suggests the chains haven't converged):
      age.65.to.79.mortality.slope.multiplier.male   age.65.to.79.mortality.intercept.multiplier.male
                                             3.949                                             2.108
                         female.to.male.multiplier     age.65.to.79.mortality.slope.multiplier.female
                                             2.013                                             1.875
             age.20.to.29.transmission.multiplier.1             age.30.to.39.transmission.multiplier.0
                                             1.864                                             1.805
             age.20.to.29.transmission.multiplier.0             age.15.to.19.transmission.multiplier.0
                                             1.755                                             1.697
                                            trate.0                                  age.assortativity
                                             1.683                                             1.668
           age.25.to.49.hiv.mortality.multiplier.0             age.20.to.29.transmission.multiplier.3
                                             1.511                                             1.426
                          log.OR.testing.intercept                                            trate.1
                                             1.392                                             1.364
                                            trate.3                                           trate.2
                                             1.361                                             1.359
            age.15.to.19.transmission.multiplier.2             age.15.to.19.transmission.multiplier.3
                                             1.325                                             1.312
            age.30.to.39.transmission.multiplier.2                                            trate.4
                                             1.282                                             1.280
 age.65.to.79.mortality.intercept.multiplier.female           age.15.to.24.hiv.mortality.multiplier.0
                                             1.277                                             1.257
            age.20.to.29.transmission.multiplier.2             age.40.to.49.transmission.multiplier.0
                                             1.248                                             1.230
          over.80.mortality.slope.multiplier.female
                                             1.226

Merged simset:
        location simulations total.weight                                                    file
 r1.lower.middle        1000         1000 simset_r1.lower.middle_step2_chains1234_2026-11-01.Rdata

Fit check (merged simset vs every calibration target, 2000-2025): 12 of 33 targets flagged
     data.type         age  sex years.with.data pct.years.in.sim.95 median.sim.vs.data last.year last.sim.vs.data
 hiv.mortality 50 and over both              26                  27              x2.93      2025            x2.81
 hiv.mortality       25-49 both              26                   0              x0.42      2025            x0.31
 hiv.mortality       15-24 both              26                   0              x2.30      2025            x1.98
 hiv.mortality       10-19 both              26                   0              x1.76      2025            x1.72
    engagement        0-14 both              11                   0          -19.9 pts      2025         -5.6 pts
   suppression        0-14 both              11                   9          -16.2 pts      2025         -0.9 pts
     awareness        0-14 both              11                   0          -14.0 pts      2025        -20.9 pts
    prevalence       15-24 both              26                  19              x1.11      2025            x1.12
     awareness         15+ both              11                   0          +2.3 pts      2025         +1.1 pts
```

r1.lower.middle step 2: likelihood

0
−10000
−20000
−30000
−40000
lue

0
25000
50000
75000
100000
iteration

chain
1
2
3
4

type
Log Likelihood
Log Prior
Log Prior x Likelihood

r1.lower.middle step 2: trace

trate.0 (1.68)

female.to.male.multiplier (2.01)

age.15.to.19.transmission.multiplier.0 (1.70)

age.20.to.29.transmission.multiplier.0 (1.75)

age.20.to.29.transmission.multiplier.1 (1.86)

age.20.to.29.transmission.multiplier.3 (1.43)

age.30.to.39.transmission.multiplier.0 (1.81)

age.assortativity (1.67)

age.65.to.79.mortality.intercept.multiplier.male (2.11)

age.65.to.79.mortality.slope.multiplier.male (3.95)

age.65.to.79.mortality.slope.multiplier.female (1.87)

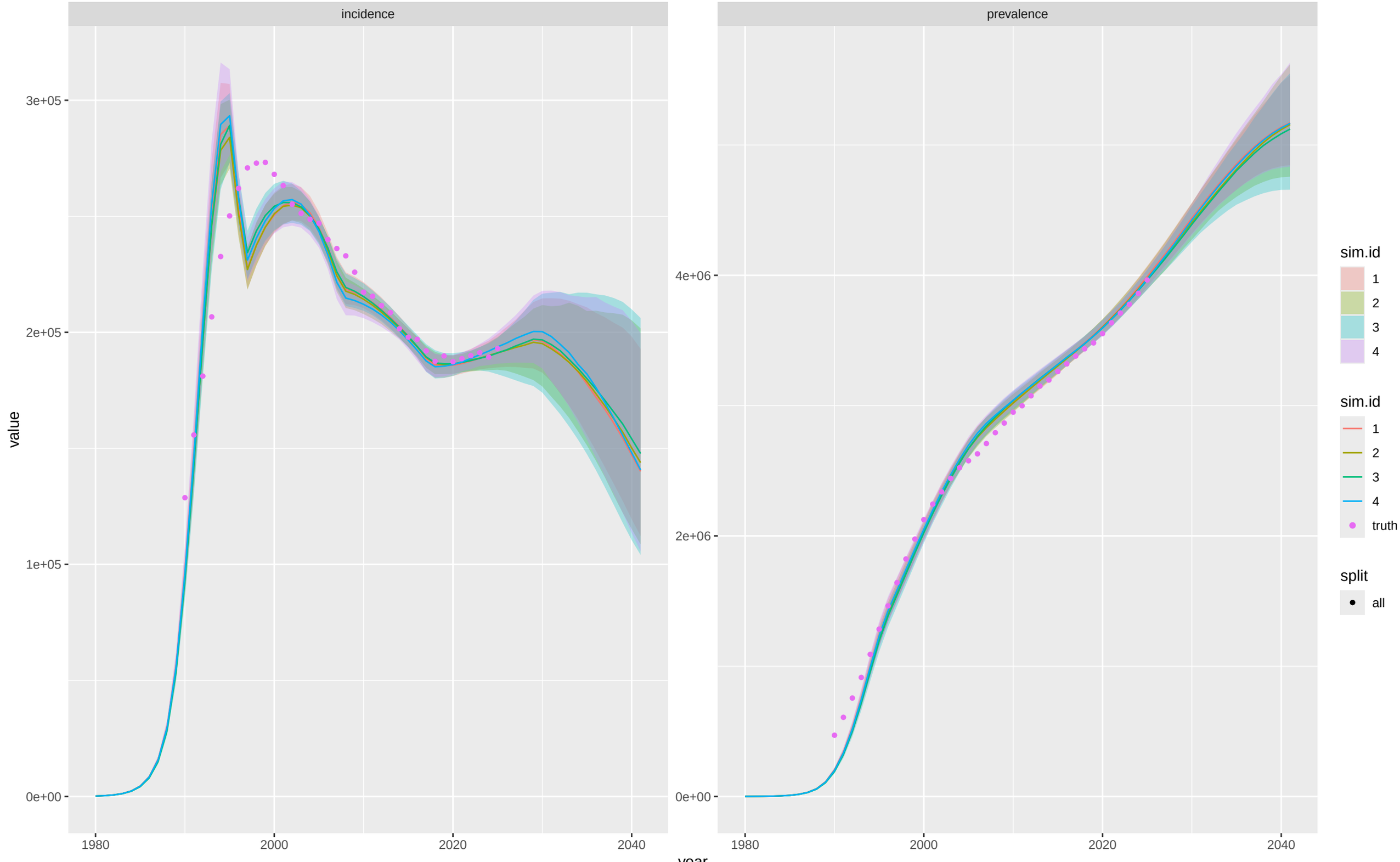
age.25.to.49.hiv.mortality.multiplier.0 (1.51)

value

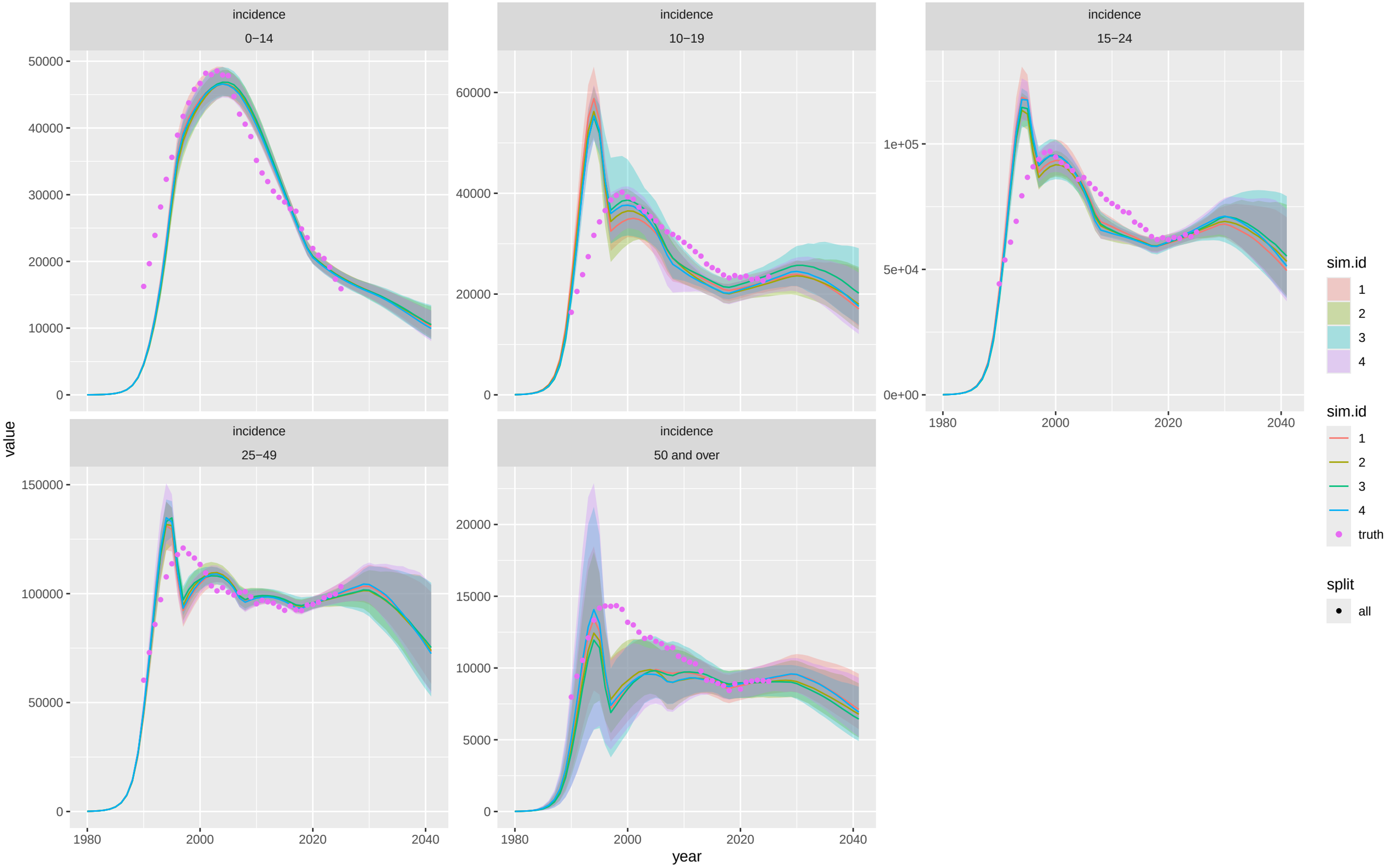
iteration

chain
1
2
3
4

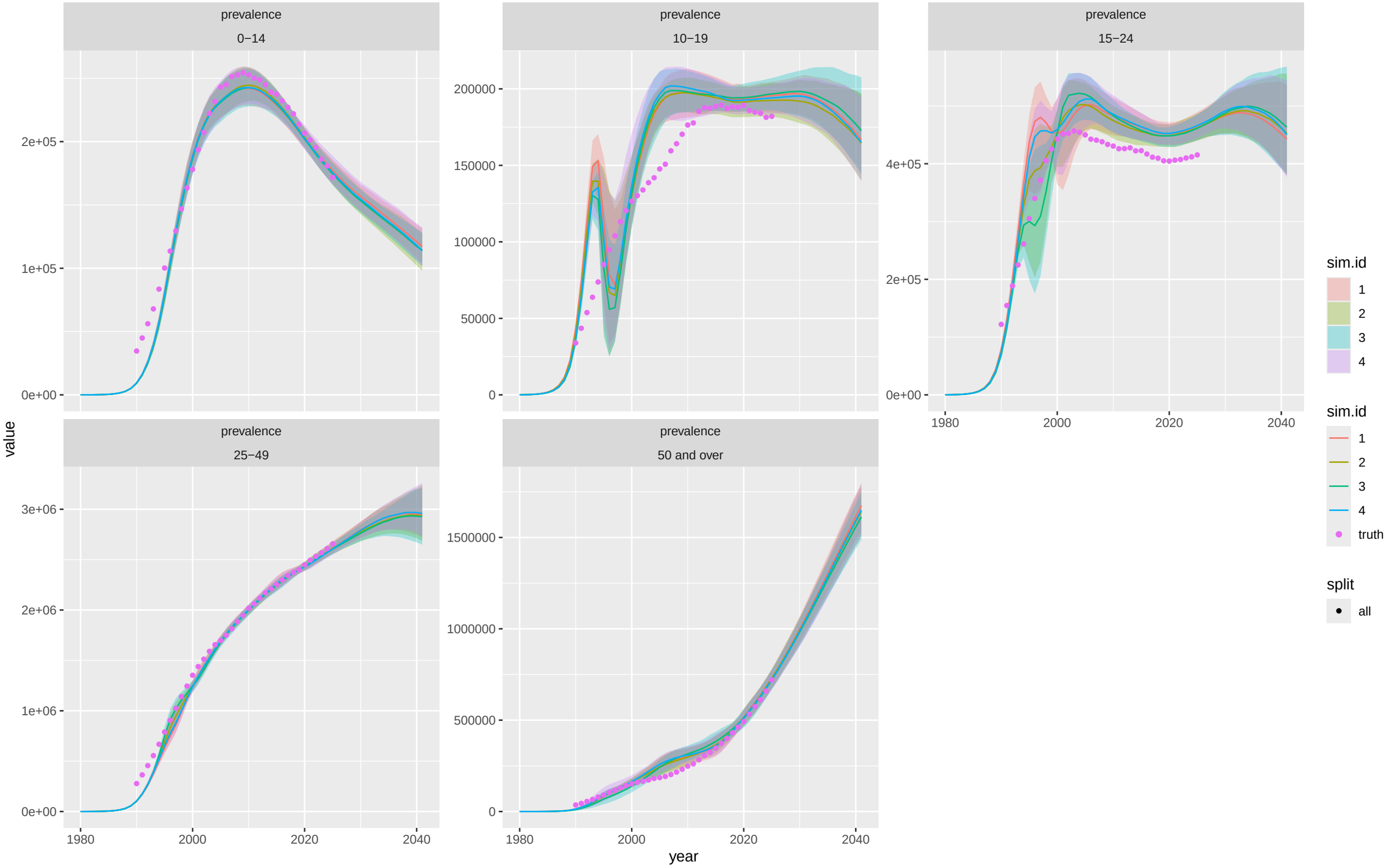
r1.lower.middle step 2: fit (one color per chain)



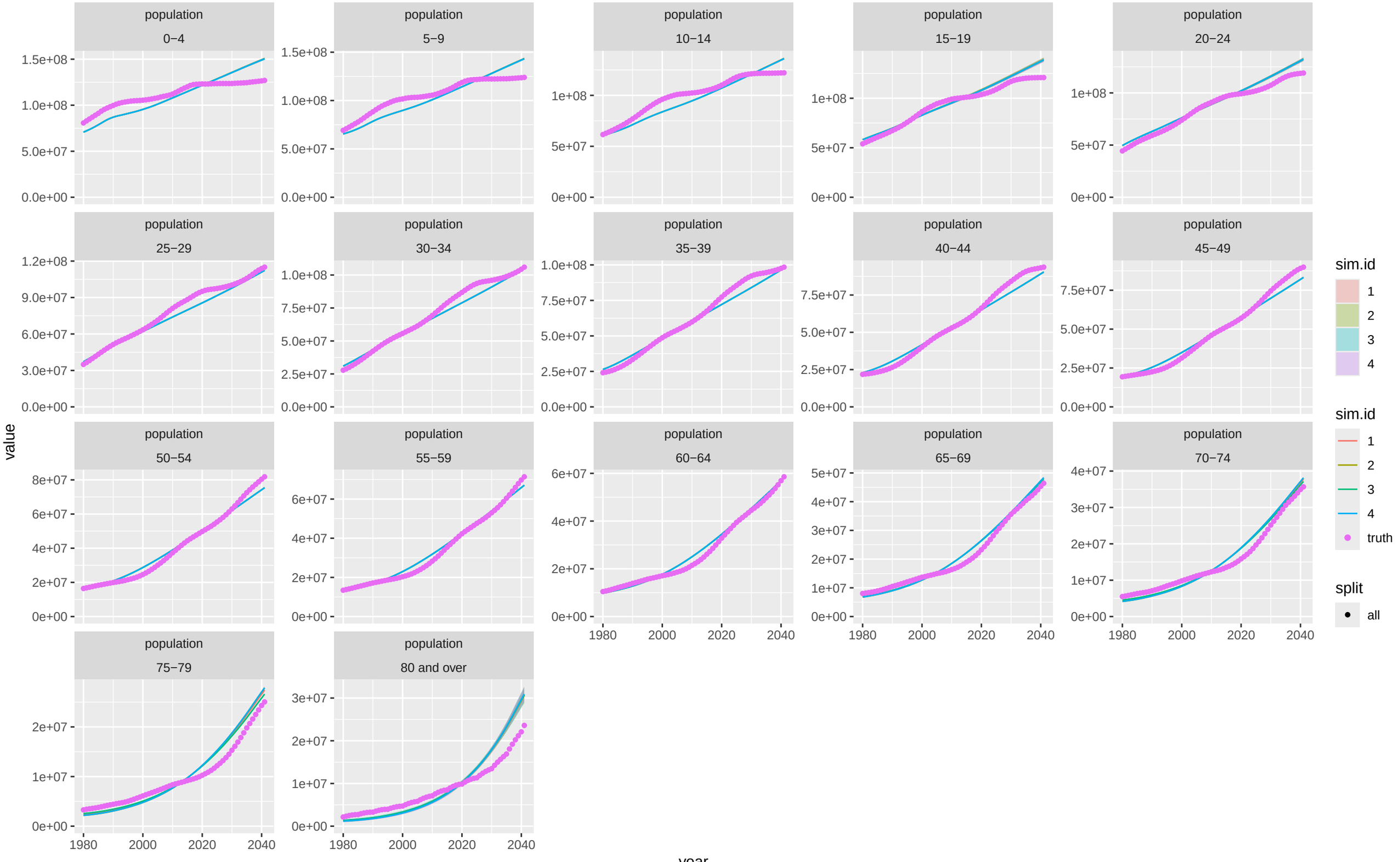
r1.lower.middle step 2: incidence.by.age (one color per chain)



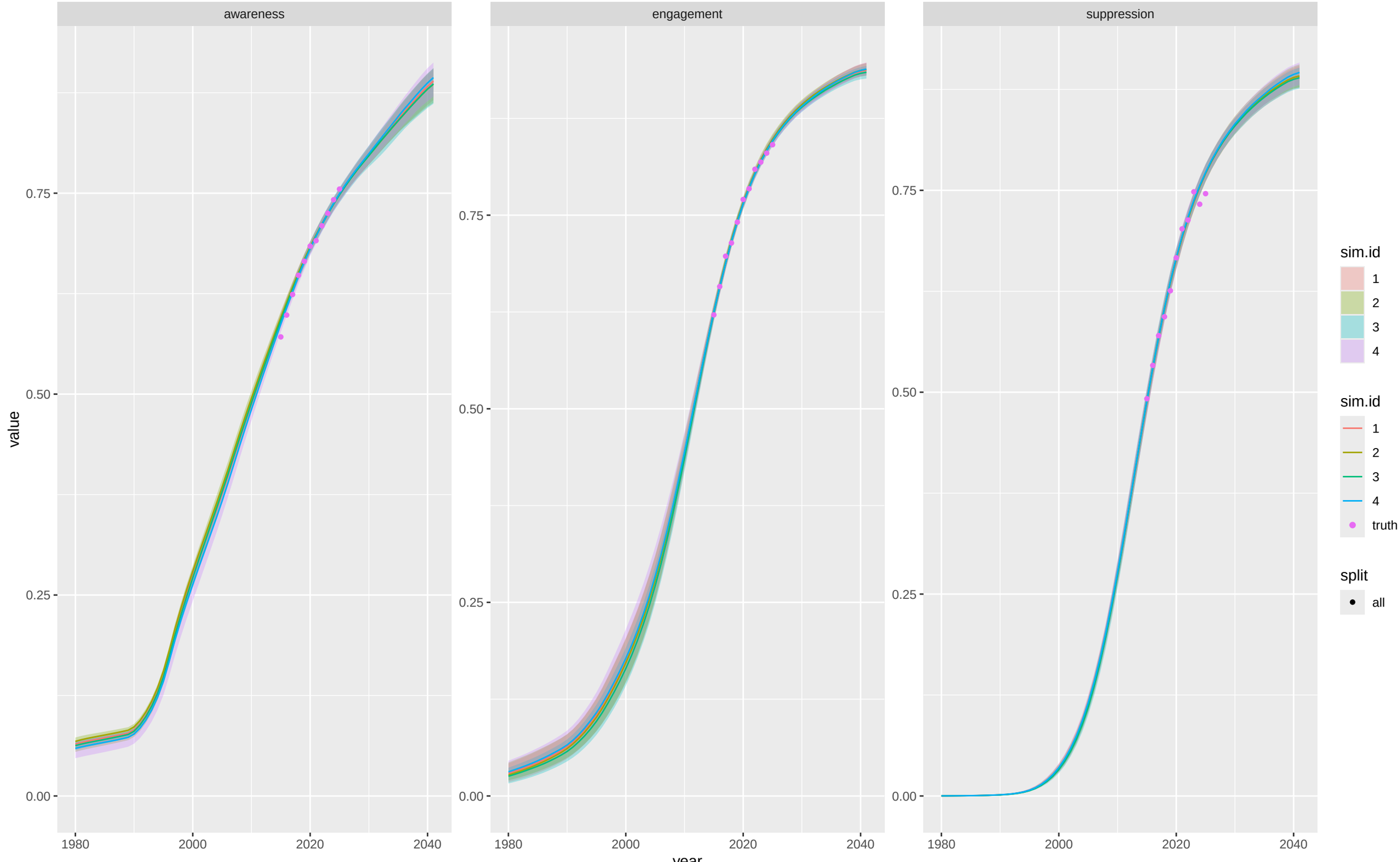
r1.lower.middle step 2: prevalence.by.age (one color per chain)



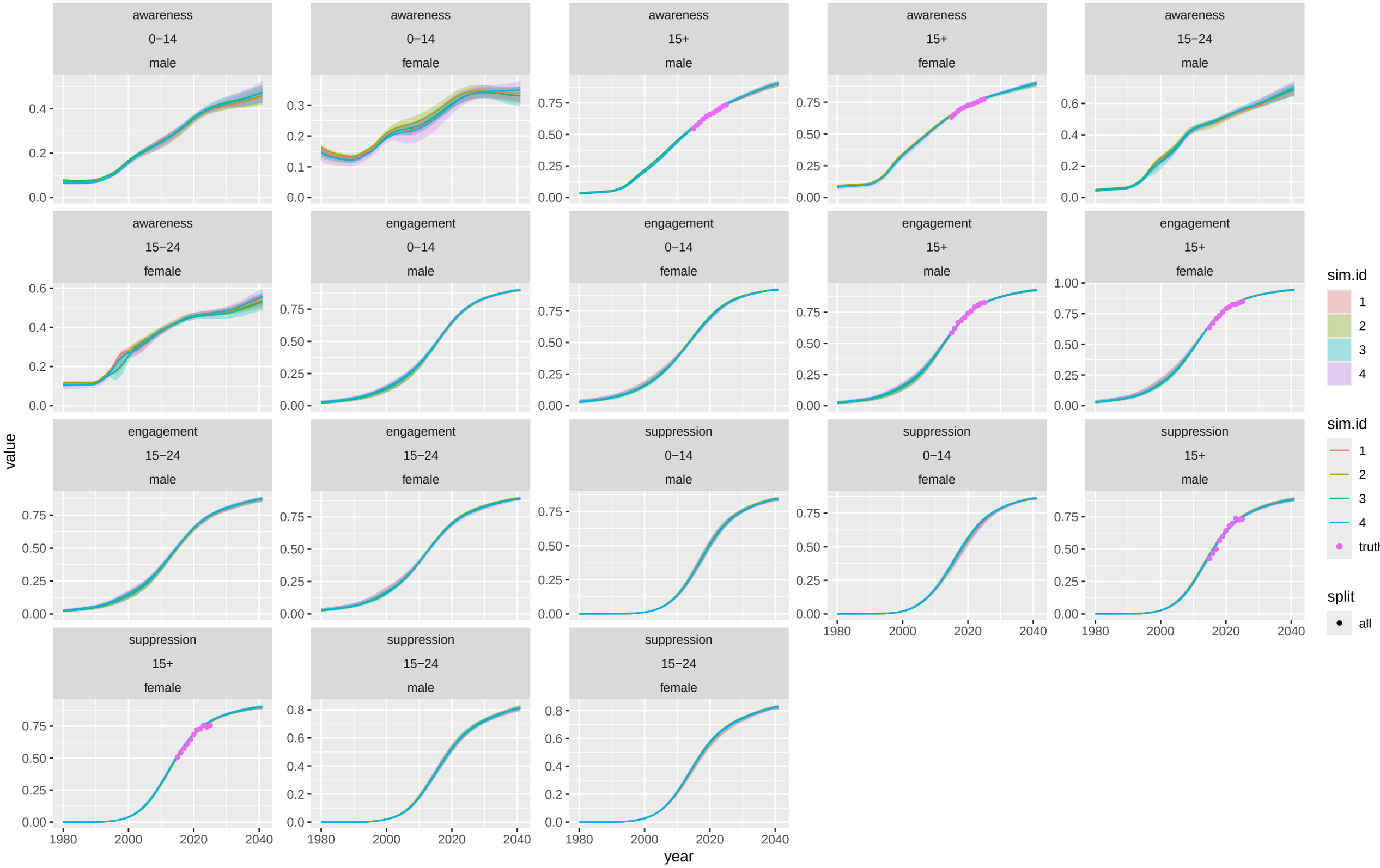
r1.lower.middle step 2: population.by.age (one color per chain)



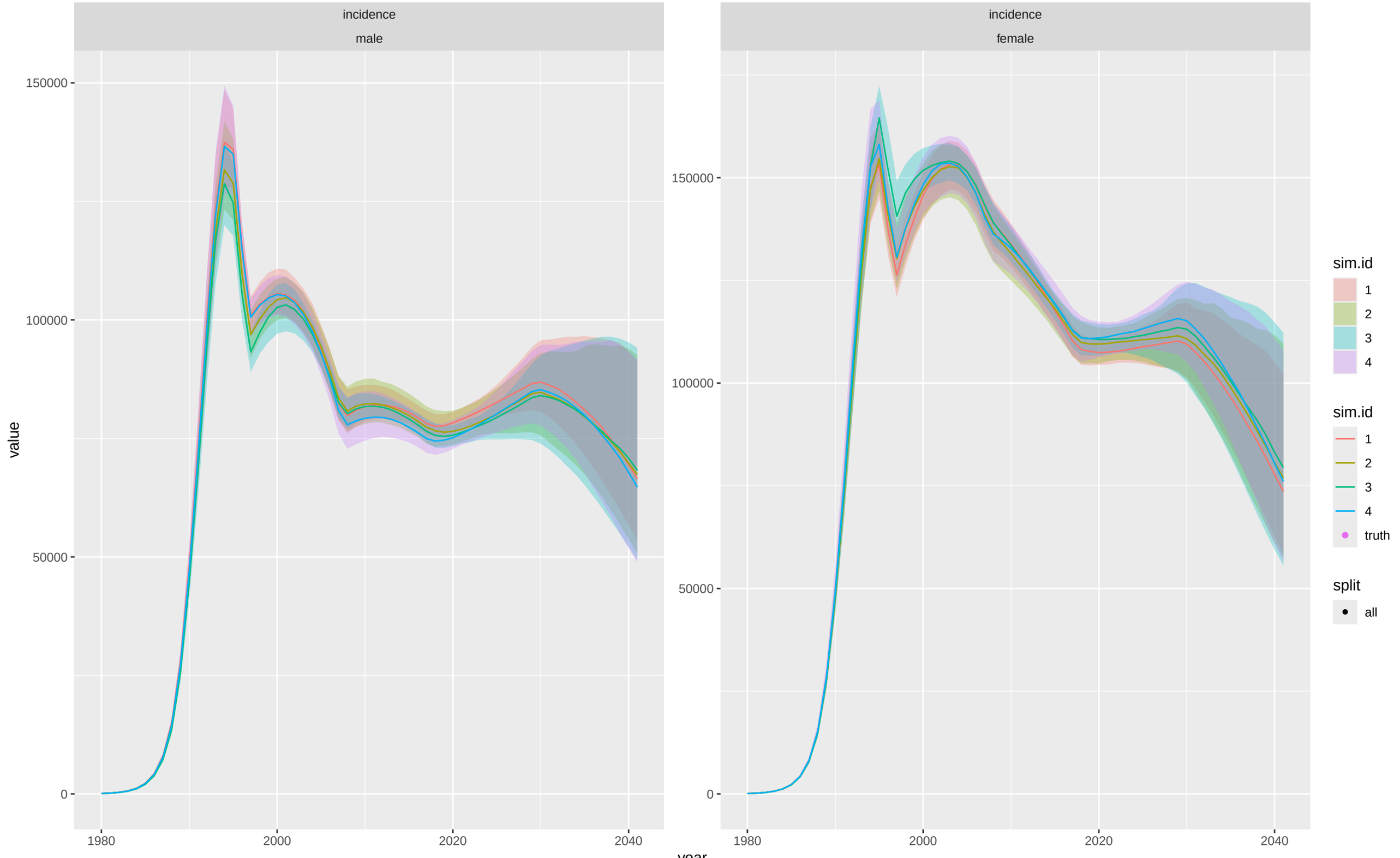
r1.lower.middle step 2: cascade (one color per chain)



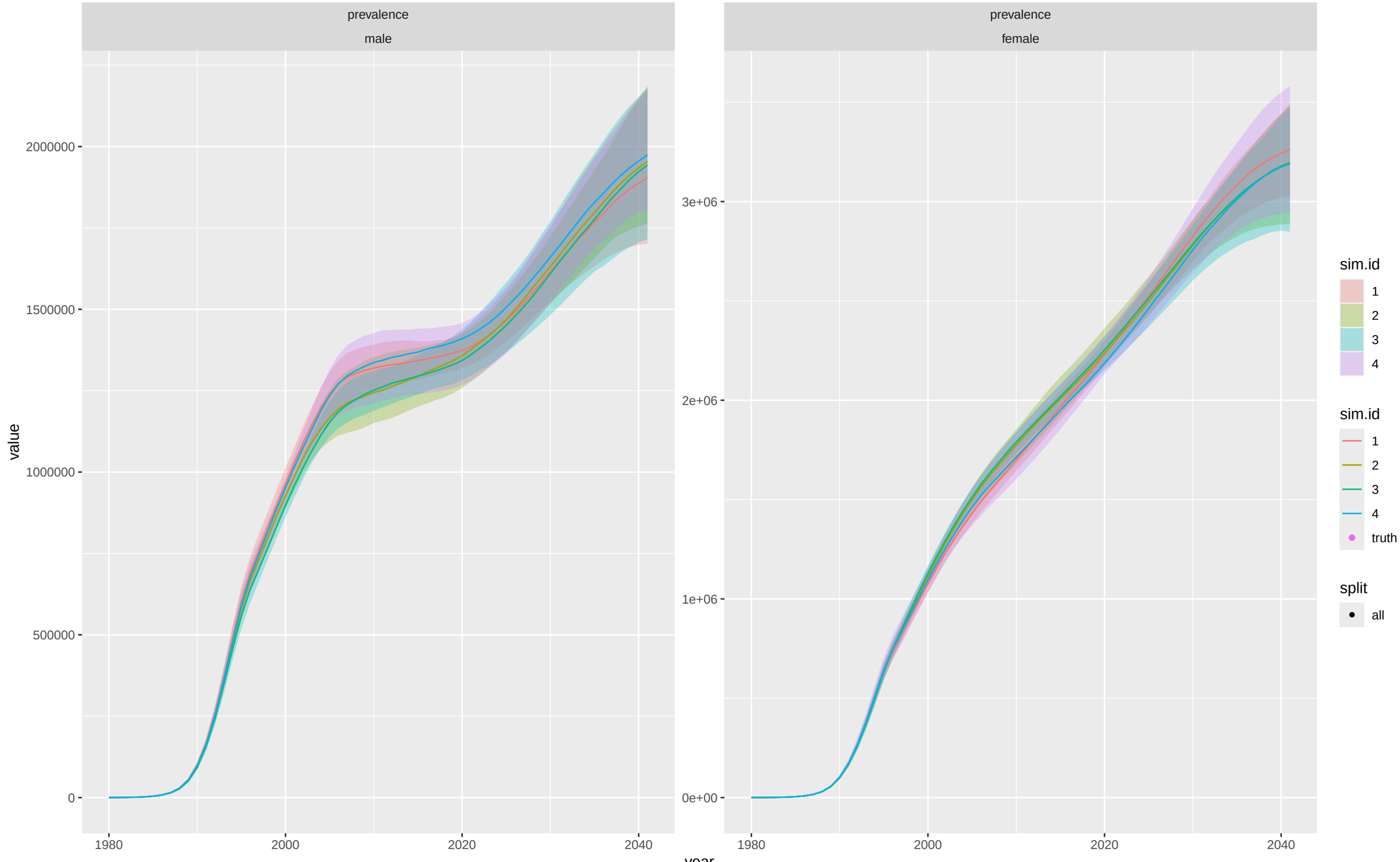
r1.lower.middle step 2: cascade.by.age.sex (one color per chain)



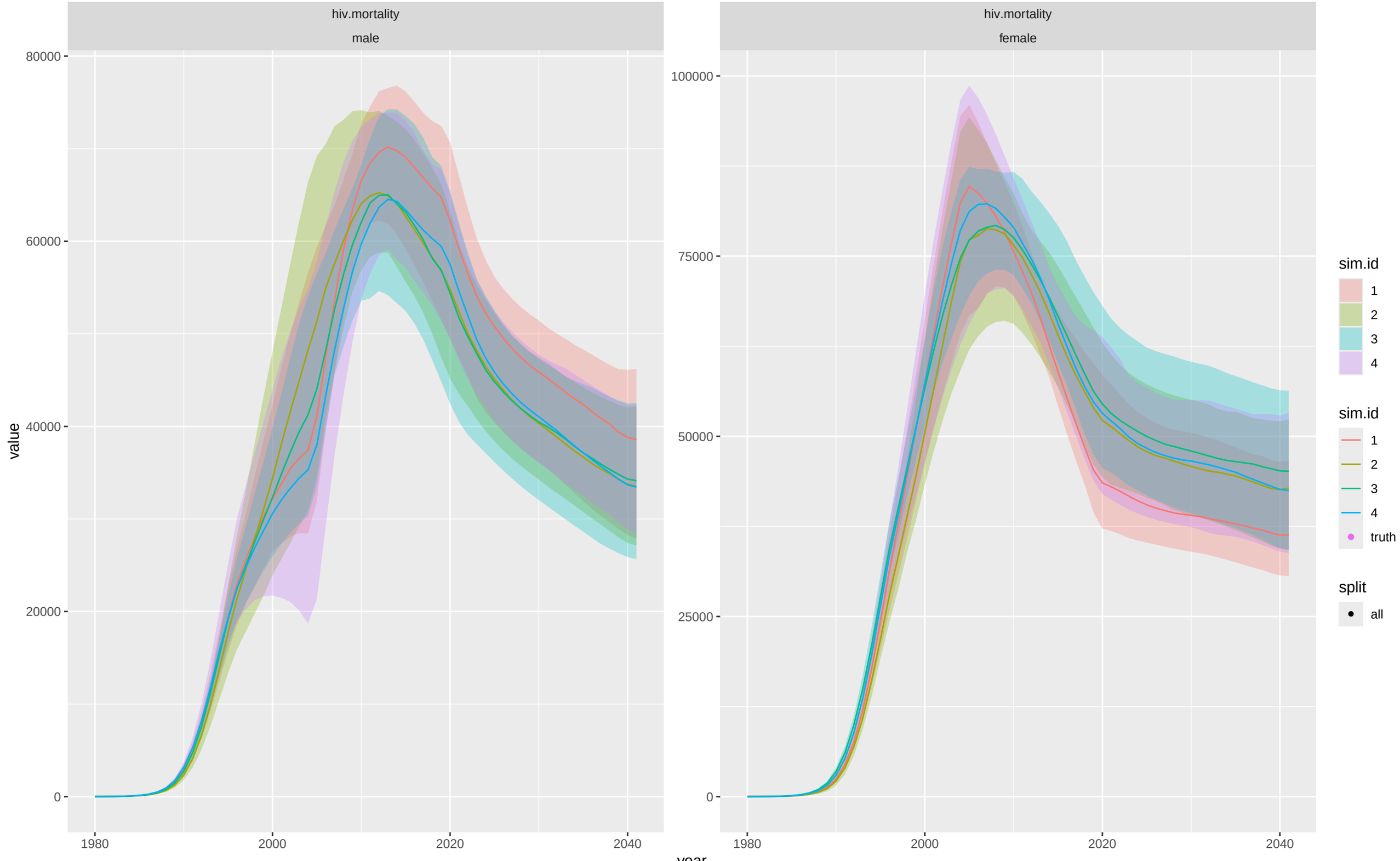
r1.lower.middle step 2: incidence.by.sex (one color per chain)



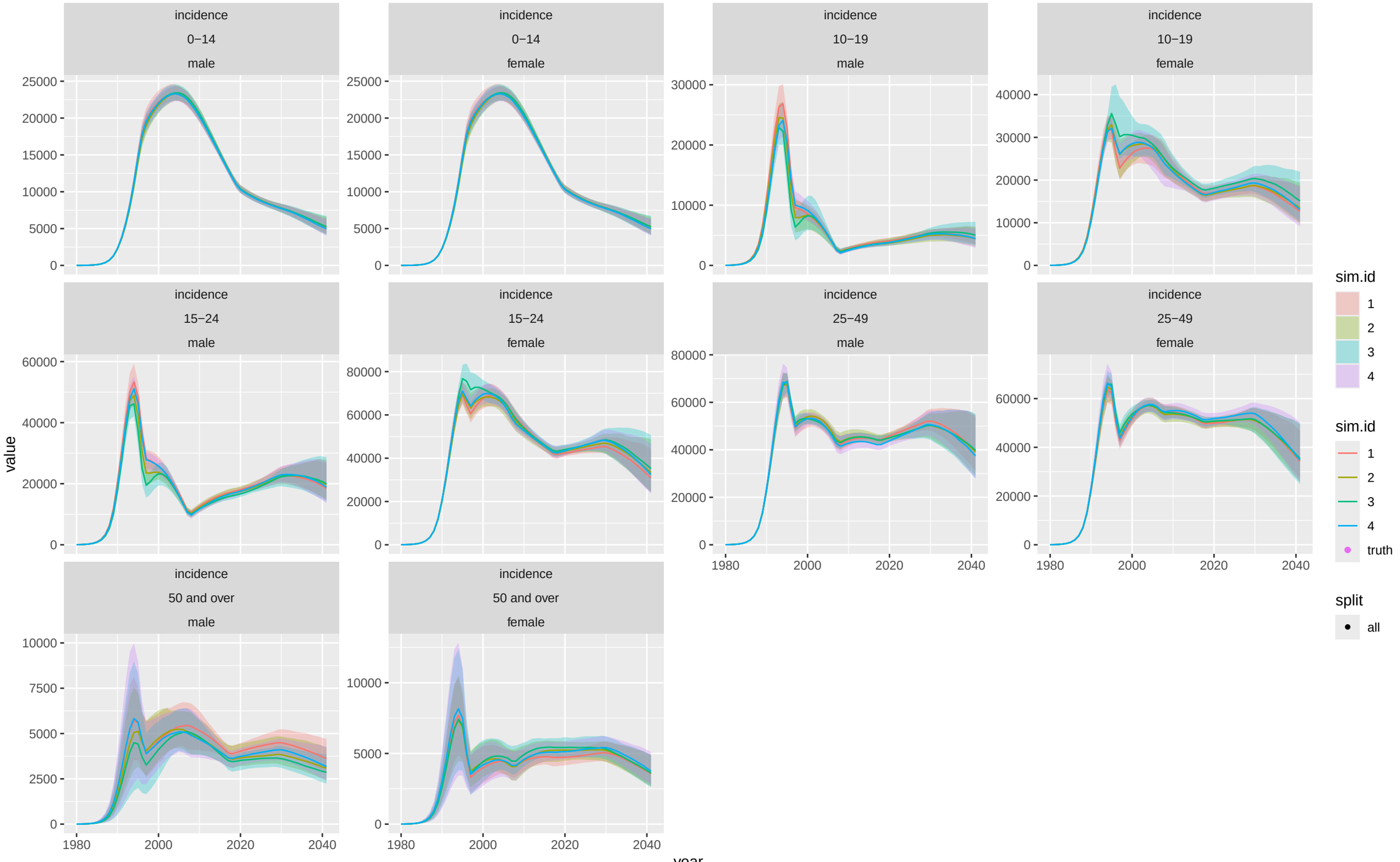
r1.lower.middle step 2: prevalence.by.sex (one color per chain)



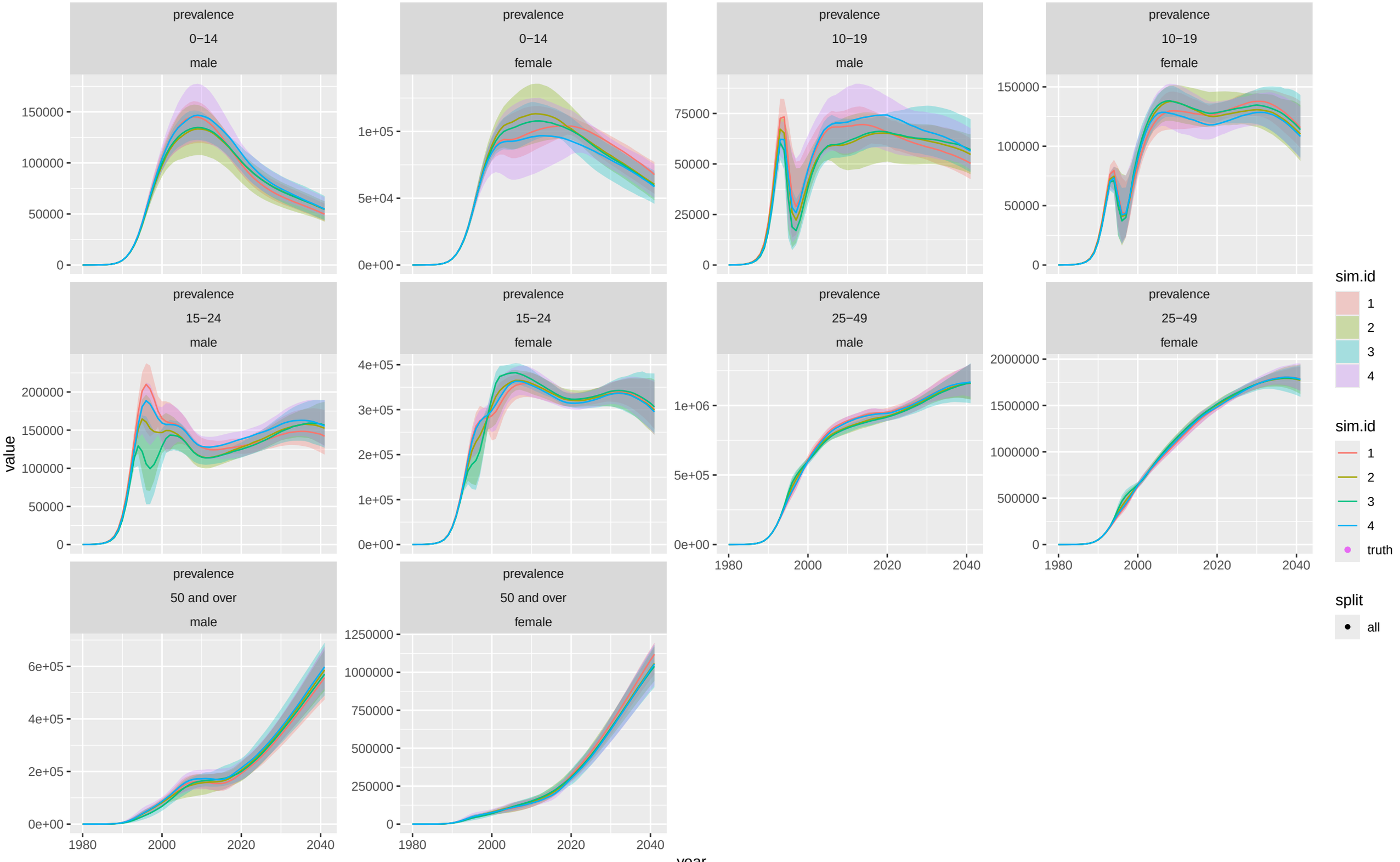
r1.lower.middle step 2: hiv.mortality.by.sex (one color per chain)



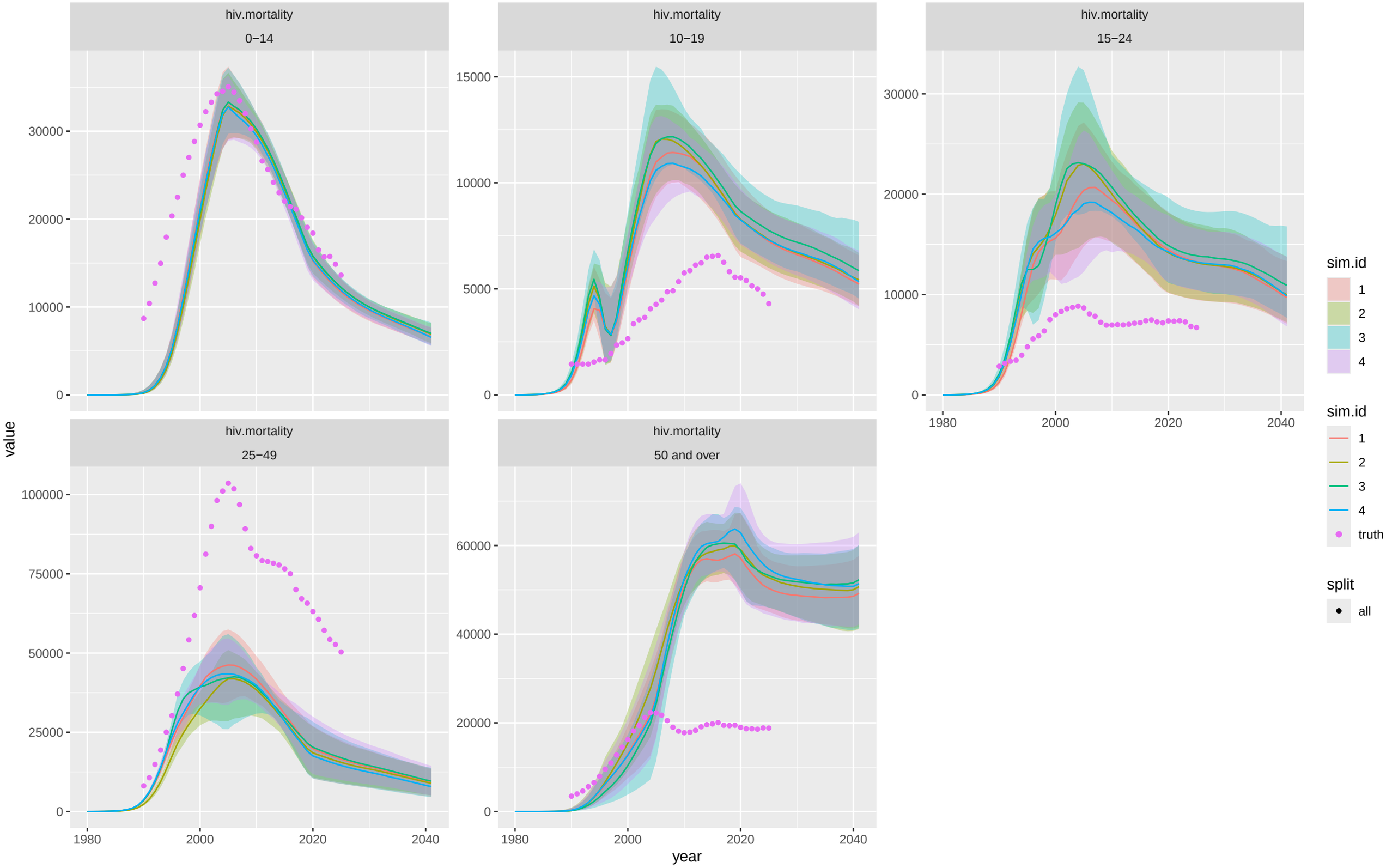
r1.lower.middle step 2: incidence.by.age.sex (one color per chain)



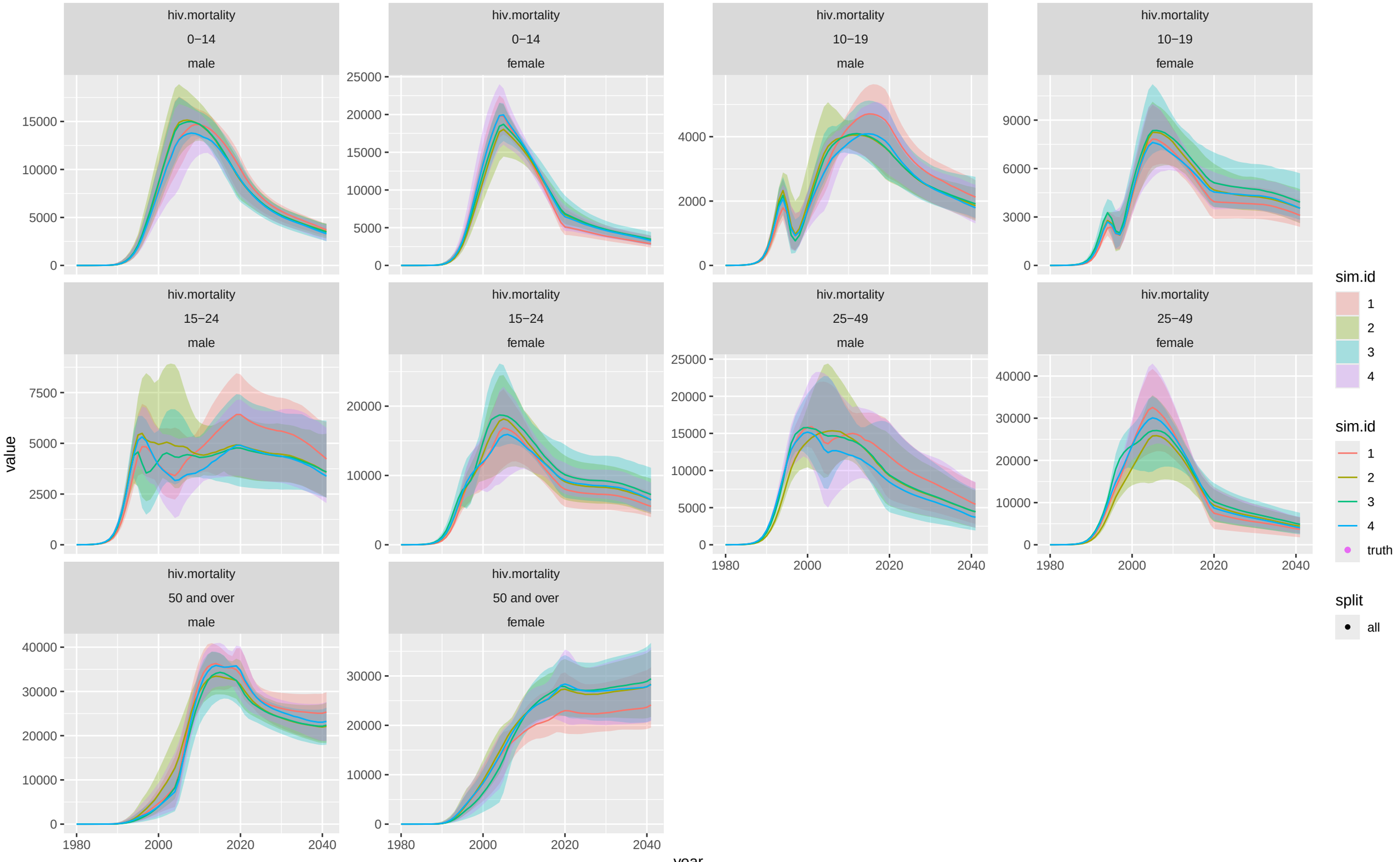
r1.lower.middle step 2: prevalence.by.age.sex (one color per chain)



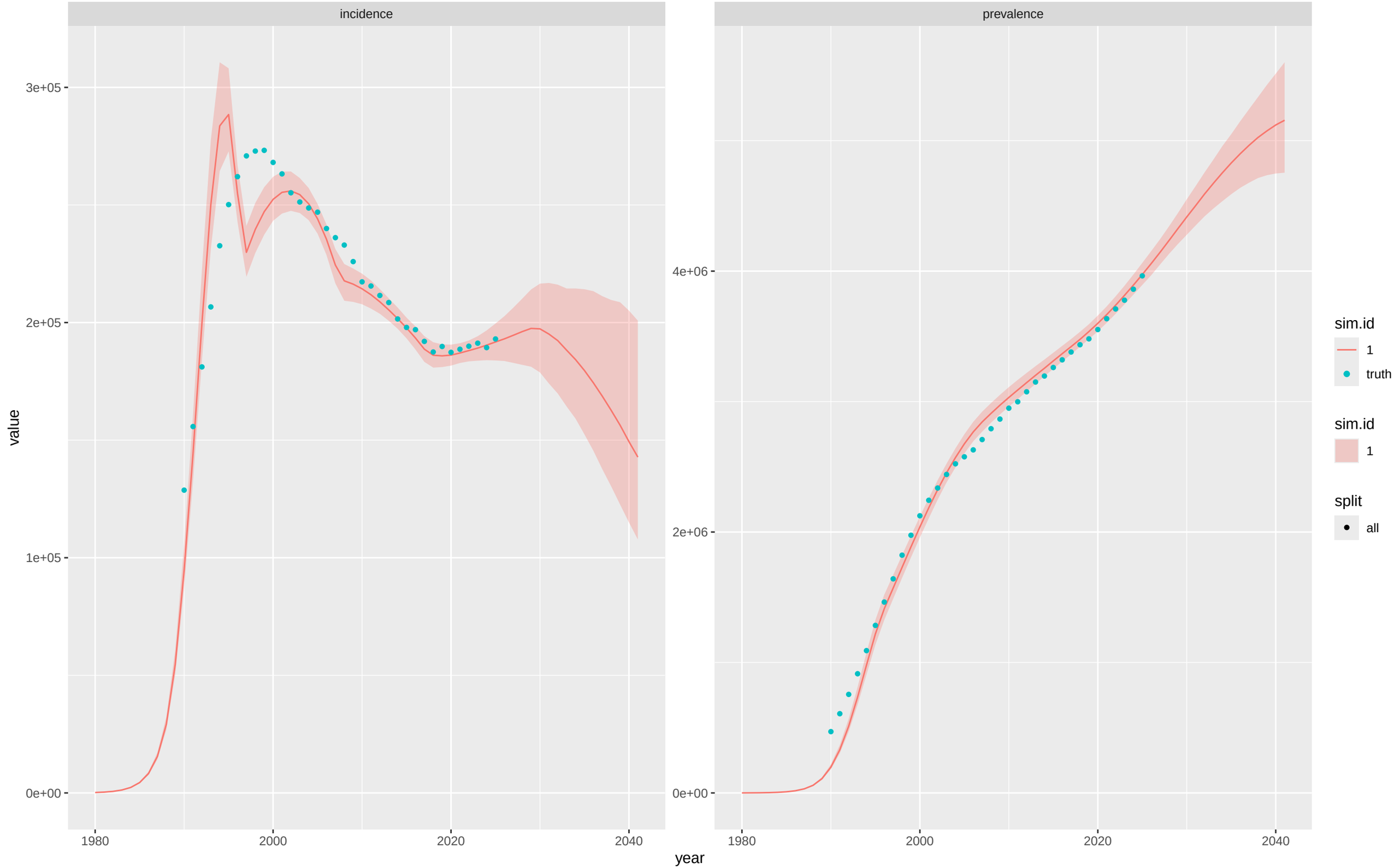
r1.lower.middle step 2: hiv.mortality.by.age (one color per chain)



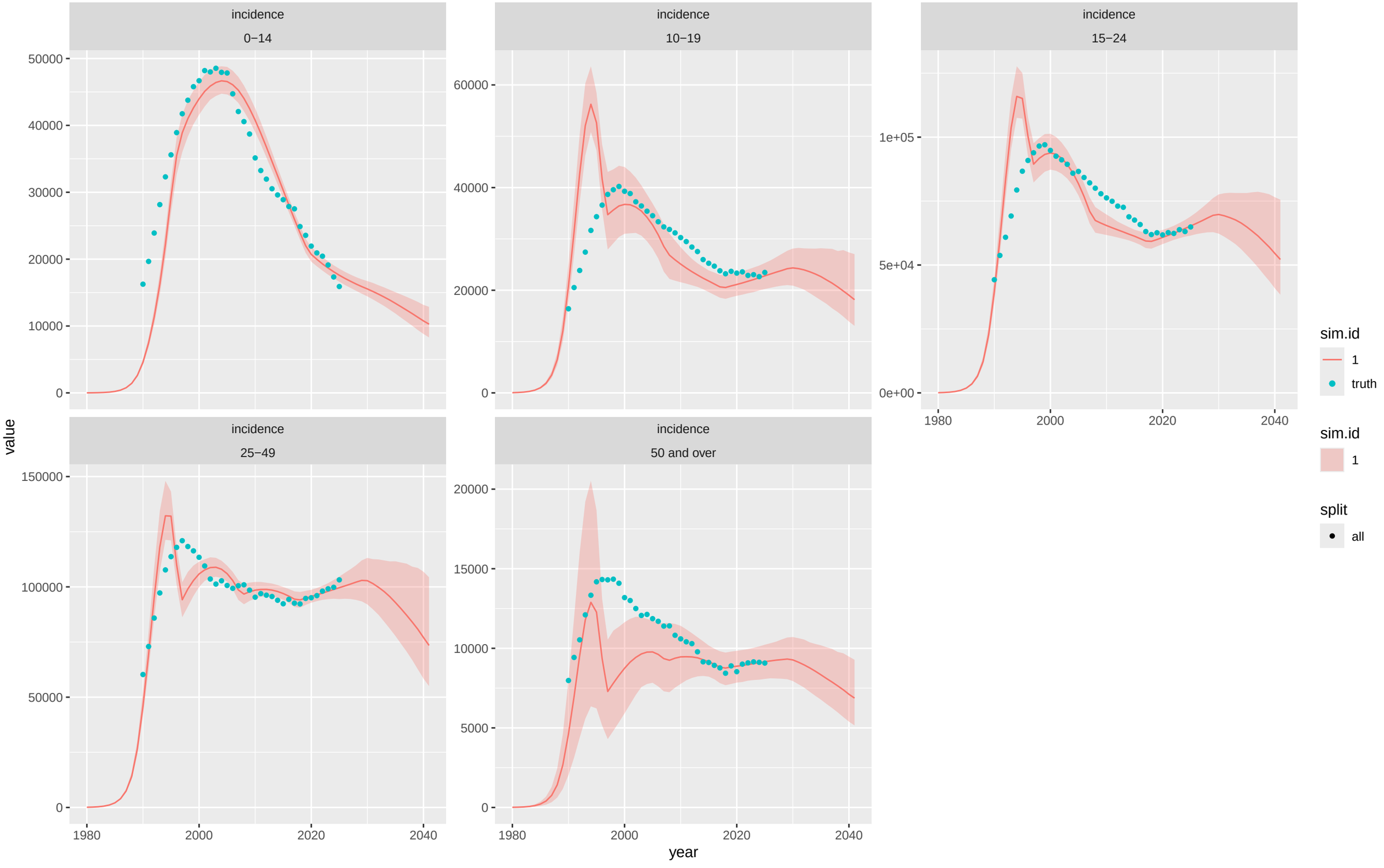
r1.lower.middle step 2: hiv.mortality.by.age.sex (one color per chain)



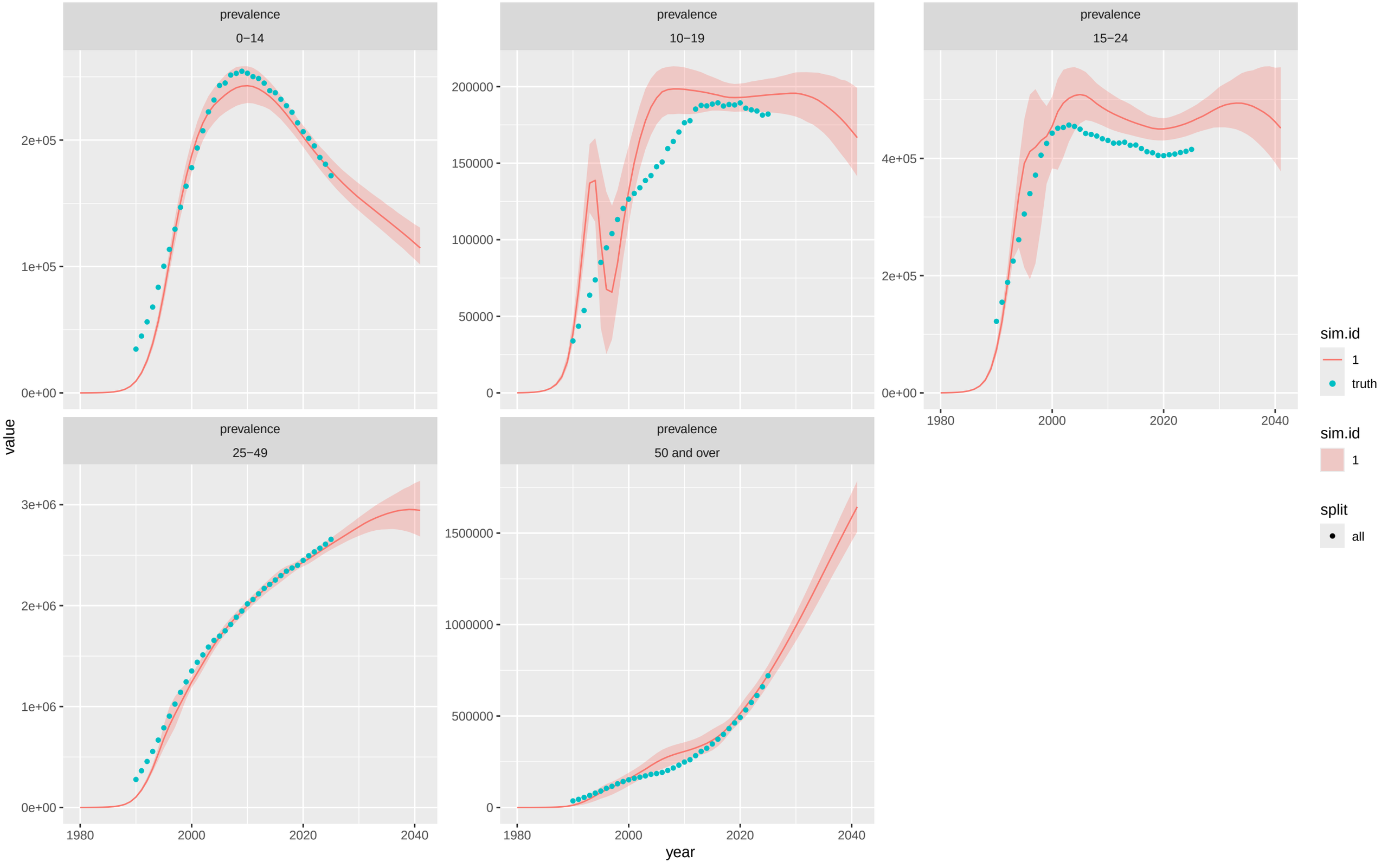
r1.lower.middle: fit (merged Step 2 simset)



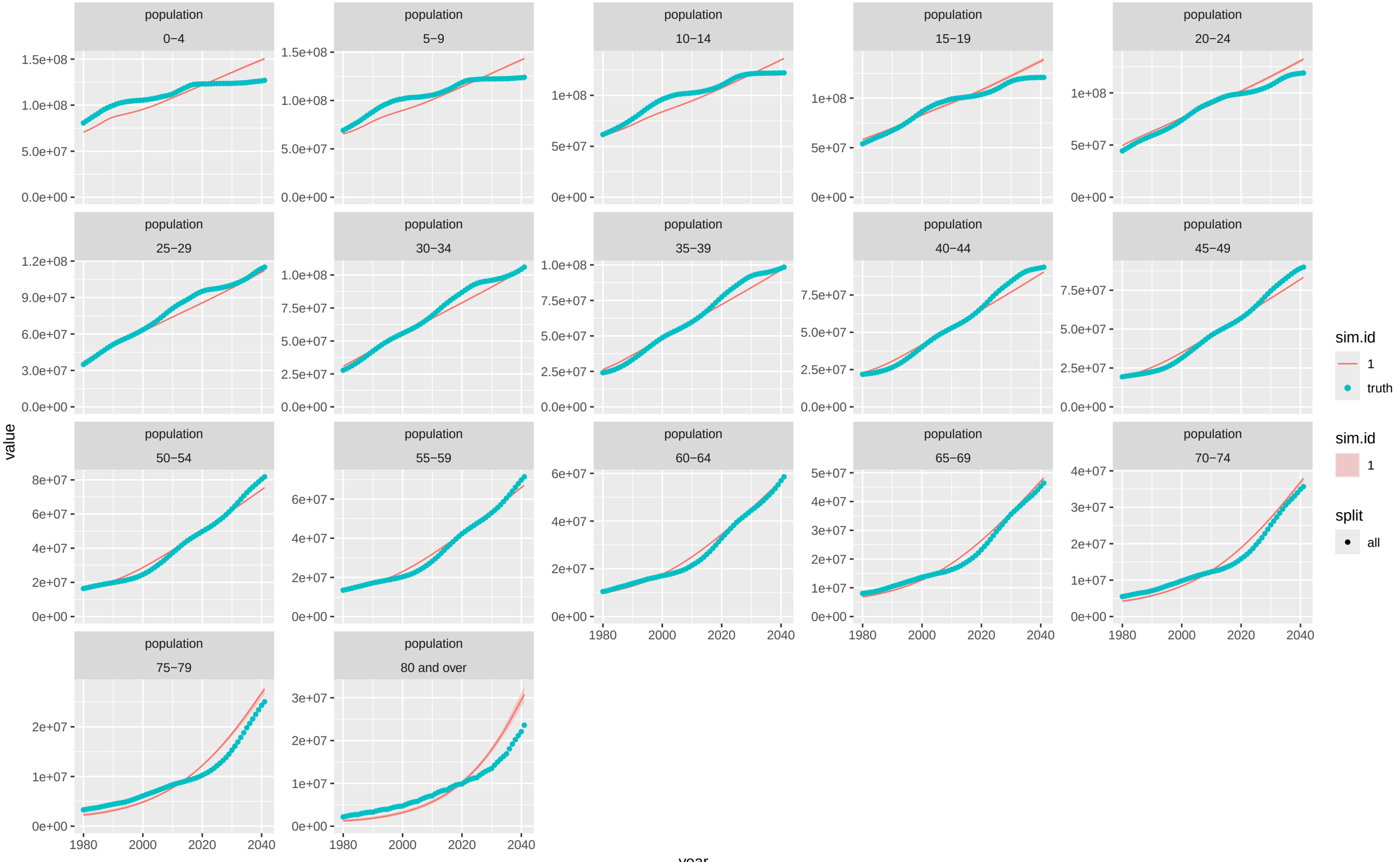
r1.lower.middle: incidence.by.age (merged Step 2 simset)



r1.lower.middle: prevalence.by.age (merged Step 2 simset)



r1.lower.middle: population.by.age (merged Step 2 simset)



r1.lower.middle: cascade (merged Step 2 simset)

awareness

engagement

suppression

value

0.75

0.50

0.25

0.00

1980

2000

2020

2040

year

sim.id

1

truth

sim.id

1

split

all

r1.lower.middle: cascade.by.age.sex (merged Step 2 simset)

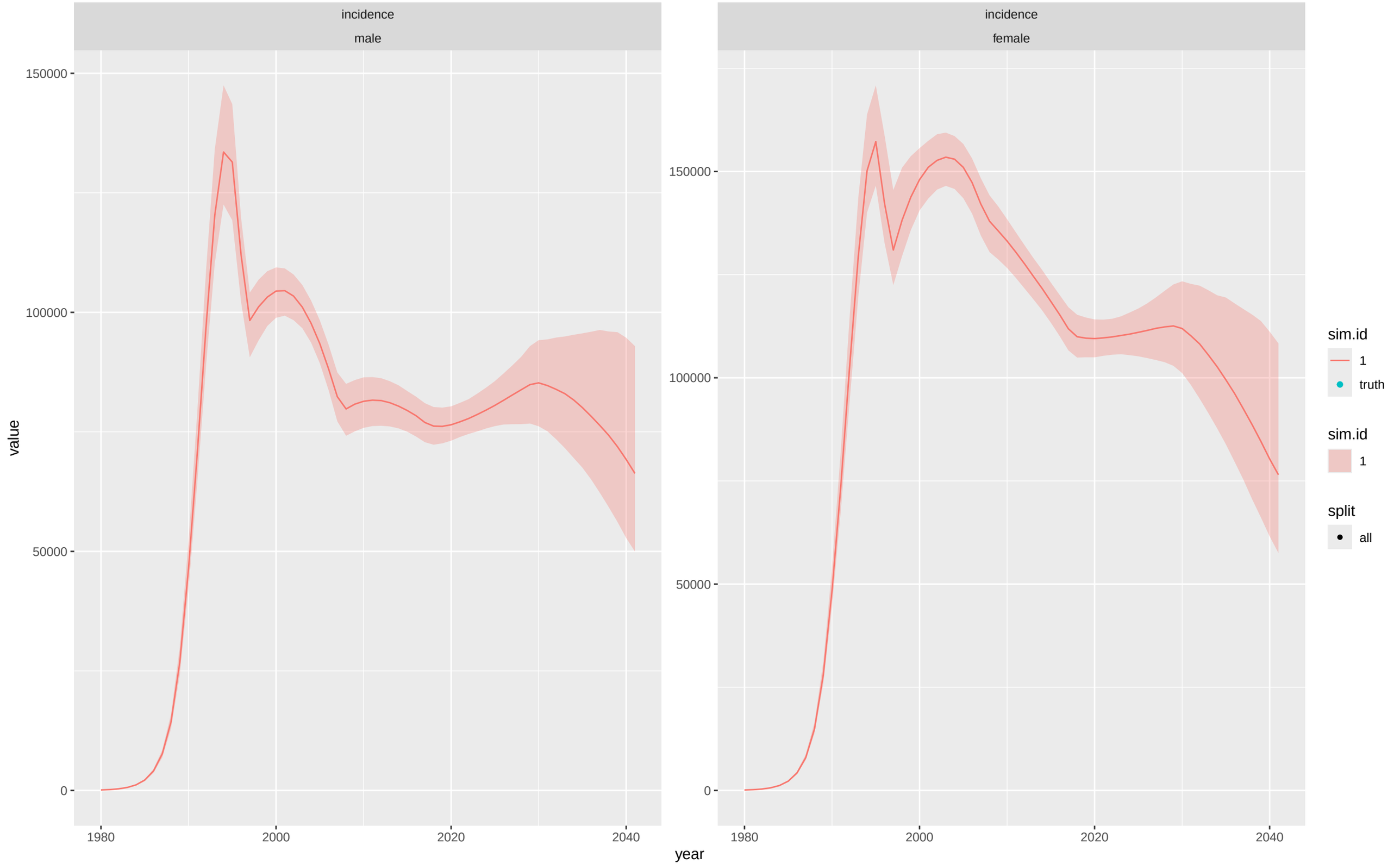
awareness 0−14 male
awareness 0−14 female
awareness 15+ male
awareness 15+ female
awareness 15−24 male
awareness 15−24 female
engagement 0−14 male
engagement 0−14 female
engagement 15+ male
engagement 15+ female
engagement 15−24 male
engagement 15−24 female
suppression 0−14 male
suppression 0−14 female
suppression 15+ male
suppression 15+ female
suppression 15−24 male
suppression 15−24 female

value
year

sim.id
1
truth

split
all

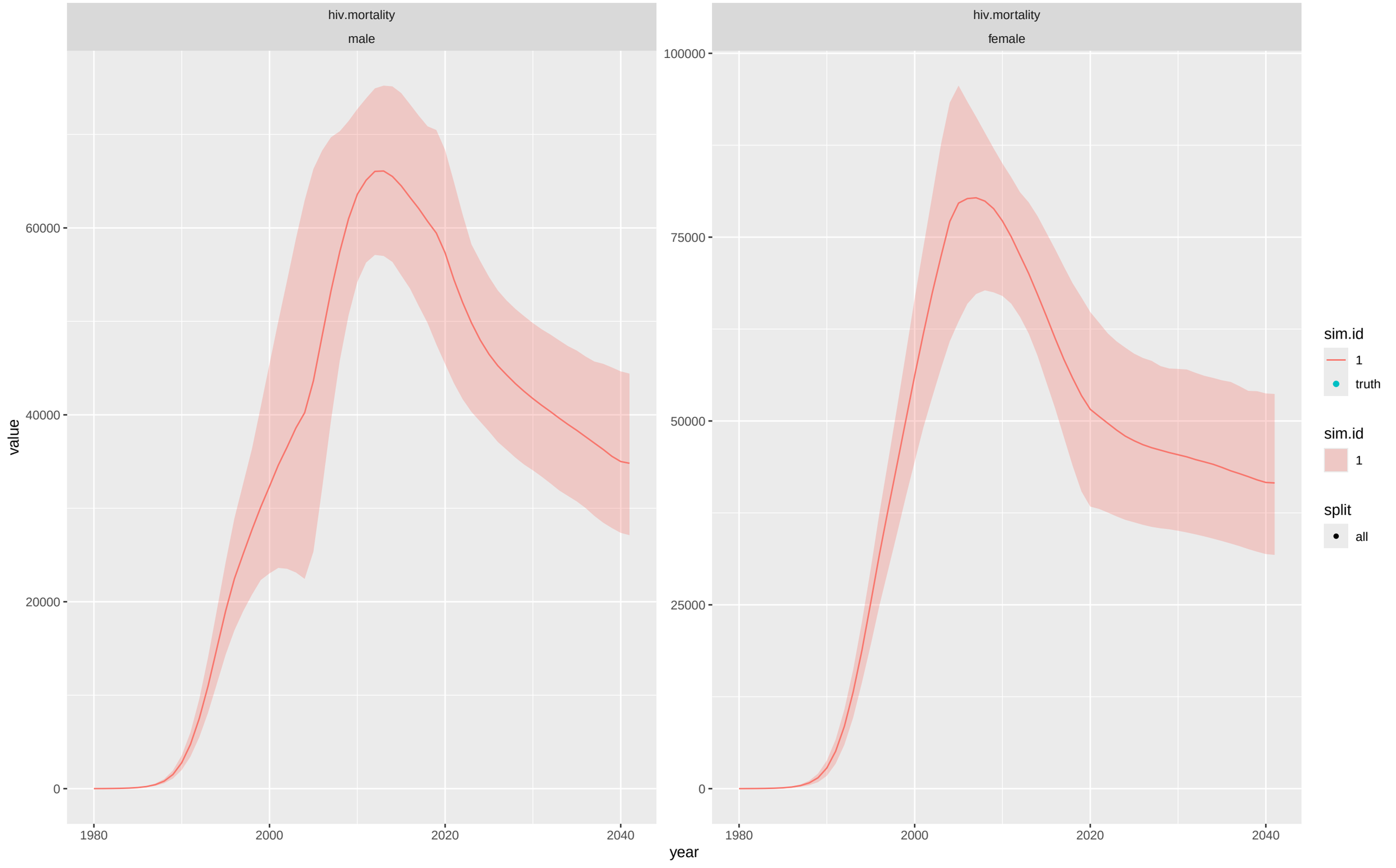
r1.lower.middle: incidence.by.sex (merged Step 2 simset)



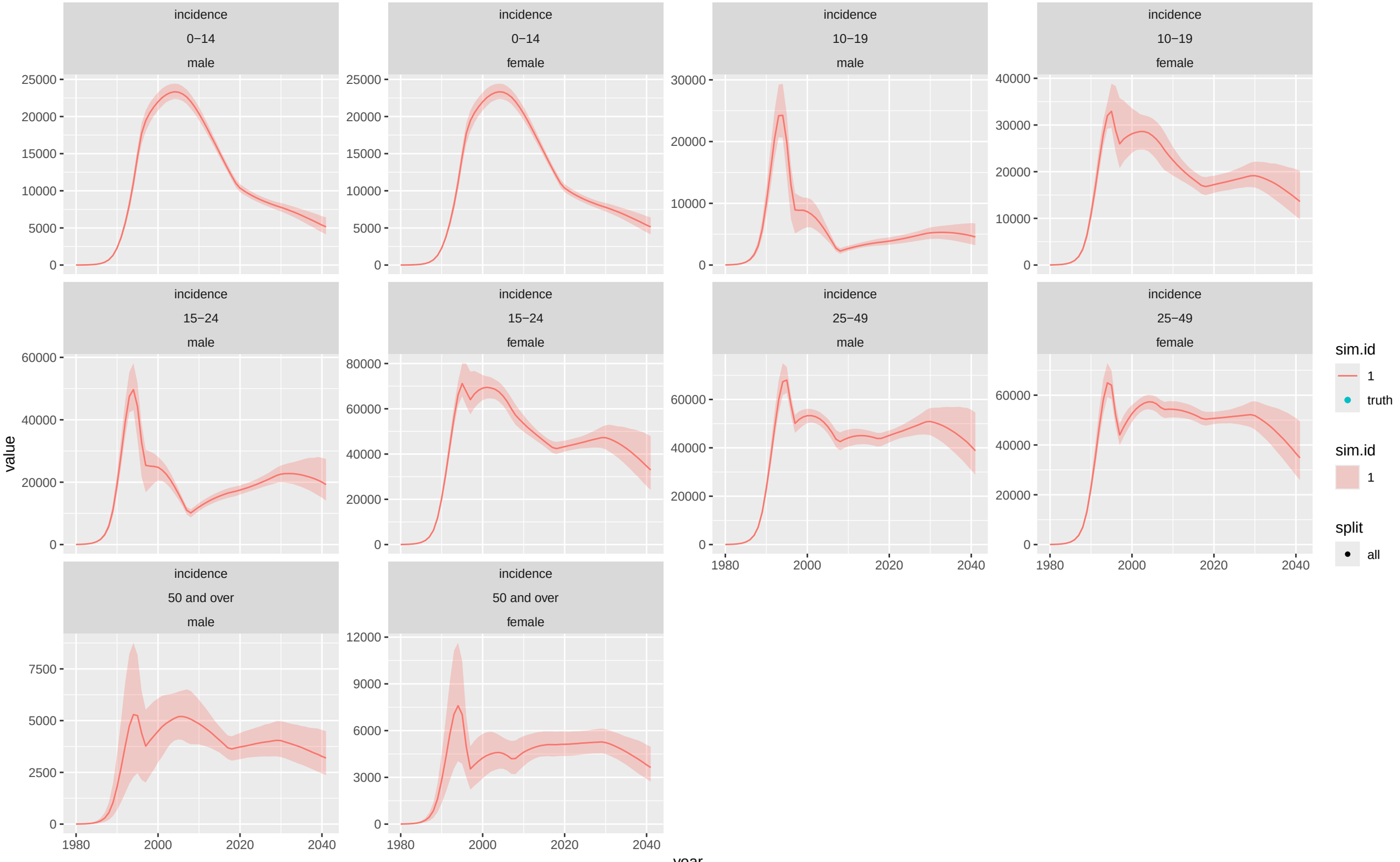
r1.lower.middle: prevalence.by.sex (merged Step 2 simset)



r1.lower.middle: hiv.mortality.by.sex (merged Step 2 simset)



r1.lower.middle: incidence.by.age.sex (merged Step 2 simset)



r1.lower.middle: prevalence.by.age.sex (merged Step 2 simset)



r1.lower.middle: hiv.mortality.by.age (merged Step 2 simset)



r1.lower.middle: hiv.mortality.by.age.sex (merged Step 2 simset)

